

Rahul Balachandar

+91 6385633609 | rahulbalachandar024@gmail.com | [linkedin.com/in/rahul-balachandar](https://www.linkedin.com/in/rahul-balachandar) | github.com/RahulB-24 | [Portfolio](#)

EDUCATION

Vellore Institute of Technology

B.Tech in Computer Science and Engineering (AI & ML) | CGPA: 8.77

Chennai, India

Aug 2023 – May 2027 (Expected)

TECHNICAL SKILLS

Languages: Java, Python, C++, SQL, TypeScript

Backend: Spring Boot, Node.js, Express, FastAPI, REST APIs

Frontend: React.js, Vite, HTML/CSS

Databases: PostgreSQL, MongoDB, Redis

DevOps & Cloud: AWS, Docker, Git, GitHub Actions, CI/CD, Linux, Vercel, Render

Machine Learning: XGBoost, LightGBM, Scikit-Learn, TensorFlow, PyTorch, Pandas, NumPy

EXPERIENCE

Software Engineer Intern

May 2026 – Jul 2026

Cimplify.AI

Chennai, India

- Built a production planning and scheduling engine that models 7 sequential manufacturing operations using batch-transfer scheduling, eliminating dependency conflicts and validating 105 component-cluster scenarios with zero workflow violations.
- Developed a 12-endpoint FastAPI backend orchestrating 6 ML models and manufacturing data pipelines to generate production schedules, procurement plans, and cost estimates from inventory, supplier, and machine-capacity data.
- Implemented inventory tracking across 7 manufacturing stages to prevent redundant scheduling, and built a scenario-analysis module that evaluates alternative machine allocations to identify bottlenecks and improve resource utilization.

AI Engineer Intern

May 2025 – Jun 2025

TANSAM

Chennai, India

- Deployed a full-stack chlorine detection system for DPD water quality testing across Tamil Nadu public health facilities, reducing manual testing time by 80%.
- Built a two-stage ML pipeline – YOLOv8 for ROI localization and ResNet-50 CNN for multi-class classification (0–5 ppm) – achieving 92% accuracy on 500+ real-world samples; served via Flask REST API with React.js frontend.

PROJECTS

ExpenseOps | Java, Spring Boot, PostgreSQL, React, Docker | [GitHub](#) | [Live App](#)

Jan 2026

- Built a multi-tenant SaaS expense platform enforcing tenant-level data isolation and JWT-based role access across 4 permission levels, featuring a 5-stage approval workflow and immutable audit trails for compliance and traceability.
- Engineered 20+ REST APIs using Spring Boot and PostgreSQL with optimistic locking for concurrent-write safety; load-tested with k6 and maintained sub-second response times under concurrent workloads.

AI SQL Optimiser | FastAPI, React, PostgreSQL, LangChain, Pinecone, Docker | [GitHub](#) | [Live App](#)

Dec 2025

- Built an AI-powered SQL optimization platform that analyzes queries, scores complexity across 4 tiers, and detects 4 SQL performance anti-patterns before generating explanations, index suggestions, and optimized rewrites through a 4-stage LangChain pipeline powered by Gemini.
- Streamed optimization results in real time using Server-Sent Events (SSE) and leveraged Pinecone vector search to retrieve similar historical query optimizations as context, improving the relevance and accuracy of LLM-generated recommendations.

LinkShelf | React, TypeScript, Node.js, Express.js, PostgreSQL | [GitHub](#) | [Live App](#)

Nov 2025

- Built a bookmark management platform with 16 REST endpoints, a normalized 5-table PostgreSQL schema, GIN-indexed `tsvector` full-text search, and a multi-browser import engine supporting Chrome, Firefox, Safari, and Edge exports with failure recovery.
- Implemented Open Graph metadata scraping with automatic timeout handling, database-level URL deduplication, and IP-based rate limiting (500 req / 15 min) to improve platform reliability and search performance.

ACHIEVEMENTS

LeetCode | **Solved 250+ Problems**

[Profile](#)

IEEE ICERECT-2025 Conference Paper Publication

[IEEE](#)

- Published in IEEE Xplore: ENDCL: Attention-Enhanced CNN-BiLSTM Model for Automated Cardiovascular Disease Detection.

HackHub'25 National-Level Hackathon Finalist (Top 150/2000+)

- Selected among Top 150 Teams from 2000+ Applicants. Built DeepShield, a Vision Transformer-based Deepfake Detection System.